🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.6C

Describe Data by Mean Absolute Deviation

Lesson 2-9 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the mean absolute deviation (MAD) to describe how spread out data is.

Language Objective: I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean Absolute Deviation | Deviation | Absolute Value | Spread | Data distribution | Variability

👆 *Tap any word to see its meaning and a picture.*

1. The average distance of each value from the mean is the ____________.

2. How far a single value is from the mean is its ____________.

3. The distance of a number from zero, always positive, is its ____________.

4. How far apart the data values are from each other is the ____________.

5. The way data values are spread out is the ____________.

6 How much the data values differ is the ______________________.

Write About the Math

Both keepers allow the same average goals. Which should the coach pick and why?

Los dos porteros permiten el mismo promedio de goles. ¿A cuál debe escoger el entrenador y por qué?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Mean Absolute Deviation** *Desviación media absoluta*
- [ ] **Deviation** *Desviación*
- [ ] **Absolute Value** *Valor absoluto*
- [ ] **Spread** *Dispersión*
- [ ] **Data distribution** *Distribución de datos*

Model: *Absolute value makes every distance from the mean positive so they don't cancel out.* — A strong answer explains deviations above and below the mean would cancel to zero, so absolute value keeps each as a positive distance.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Both keepers have a mean of ____. Keeper A's MAD is ____, which means ____. Keeper B's MAD is ____, which means ____. The coach should pick ____ because ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean Absolute Deviation
2. **Notes 2:** Deviation
3. **Notes 3:** Absolute Value
4. **Notes 4:** Spread
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Mean Absolute Deviation is adding the signed deviations (value − mean) instead of their absolute values, so the positives and negatives cancel out to 0. For 6, 8, 10, 12 (mean = 9), the deviations are −3, −1, 1, 3 — they sum to 0, but that is NOT the MAD. Before averaging, take the absolute value of every deviation so each becomes a positive distance: |−3| = 3, |−1| = 1, |1| = 1, |3| = 3. Sum = 8, so MAD = 8 ÷ 4 = 2. A MAD of exactly 0 is a warning sign — it only happens if every value in the data set equals the mean.*
8. **Try It 1:** B. 2 — *Add the distances: 3 + 1 + 1 + 3 = 8. Divide by the 4 games: 8 ÷ 4 = 2. The MAD is 2 points.*
9. **Try It 2:** C. Shooter A, because a smaller MAD means scores stay closer to the average — *A smaller MAD means the scores hug the mean. Shooter A's MAD of 2 is smaller than Shooter B's 6, so Shooter A is more consistent.*